machine may be in a condition that they need to be replaced soon. To correct all such difficulties, the machine is stopped from functioning to do all and every repairs to bring back the machine to accurate machining conditions. This called as capital repairs (or) corrective maintenance .

6.18 Planned maintenance programme

The main objective of planned maintenance programme is to increase the production by keeping the machine tools always ready in good condition. It is done by keeping the machine tools to perform all the activities correctly with the required accuracy at desired speed with full safety protection.

The planned maintenance programme is to be prepared and executed as follows.

1. When a new machine tool arrives to the machine shop or the existing machine tools are to be replaced, it is required to install the machine tools properly, level and align them correctly and connect them to the electrical terminals safely.
2. If errors are found in the dimensional accuracy, the errors should be recorded and analysed whether the machine is in bad condition.
3. It is necessary to plan and get ready the materials, spares and tools required for the maintenance in advance so that the maintenance work is carried out in time.
4. The operators and supervisors should know the importance of the cutting speed, feed data of the machine tools.
5. Emergency repair works should be done without any delay in the case of breakdown of any of the machine tools.
6. The machine should be overhauled if the working efficiency of the machine tool goes below a particular level.
7. If the machine tools become very old and not performing to the desired level, it has to be dismantled completely and worn out parts should be replaced to bring it to the normal working condition.
8. The following details should be prepared and made as charts the layout plan of the shop, the size and specification of the machine tools and the parts to be lubricated. The manuals and the list of spare parts of all the machine tools are also to be kept ready.
9. Annual budget for the maintenance work should be prepared atleast six month or once in year in advance.

6.19 Instruments needed for maintenance

1. Steel rule
2. Try-square
3. Calipers
4. Micrometer
5. Vernier calipers
6. Gauges
7. Vise
8. Files
9. Tap & Die
10. Screw gauges
11. Spanner set
12. Hammers
13. Screw drivers
14. Wrenches
15. Hacksaw frame & blades
16. Lubricating devices
17. Emery sheets
18. Scraper

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